ACADEMIC RESEARCH PROJECT

Deep Learning for Multi-Stage Alzheimer's Disease Detection from Brain MRI

Natalia Iglesias Bueno, Sandra Millán, Ángel Pardo | 2023-2024

PROJECT DETAILS

- ***Role*** → Deep Learning & Biomedical Computer Vision Researcher
- ***Supervision & Setting*** → Universidad Francisco de Vitoria (UFV Madrid)
- ***Technical Stack*** →
  - **Core & Analytics:** Python (NumPy, Pandas, SciPy, Scikit-Learn)
  - **Deep Learning & Computer Vision:** TensorFlow, Keras, ResNet-50 (Transfer Learning), OpenCV
  - **Data Visualization:** Matplotlib, Seaborn
- ***Core Domains*** →
  - Medical image preprocessing & standardization (matrix normalization, channel resizing)
  - Dataset rebalancing & class-weighted loss optimization
  - Deep convolutional transfer learning & architecture benchmarking
  - Regularization & early convergence tuning (loss/accuracy curve optimization)
  - Multi-class performance auditing (confusion matrices, probability calibrations)
  - Model explainability & saliency gradient mapping (pixel attribution, anatomical localization)
  - Human-in-the-loop clinical workflow integration

CLINICAL CONTEXT & PROBLEM STATEMENT

Alzheimer's Disease is the leading cause of dementia worldwide, causing progressive memory loss and physical shrinkage of brain tissue. While brain Magnetic Resonance Imaging (MRI) is routinely used to spot structural damage, traditional clinical evaluation faces key challenges:

- **Manual and Late Diagnosis** → Visually inspecting MRI slices is time-consuming, subjective, and often catches the disease only in advanced stages—when therapeutic interventions are far less effective.
- **Subtle Early Transitions** → Telling the difference between normal aging, mild cognitive decline, and early Alzheimer's is very hard by eye because early anatomical changes are minimal and easy to miss.
- **The Black-Box Barrier** → While AI models can classify scans quickly, doctors cannot trust automated decisions unless they can clearly see *which* brain structures the algorithm is evaluating.

The objective

Main goal

Build an explainable deep learning pipeline to automatically classify brain MRIs into four progressive stages of Alzheimer's, enabling faster, earlier, and more consistent diagnosis.

Key Steps:

1. ***Data Cleaning & Balancing****: Standardize brain MRI scans and balance class distributions across all stages.*
2. ***Transfer Learning****: Train a pre-trained ResNet-50 network to recognize multi-stage disease patterns.*
3. ***Convergence Optimization****: Tune training epochs and loss curves to prevent overfitting on medical images.*
4. ***Visual Auditing****: Use gradient saliency maps to verify that the model relies on real anatomical markers (such as enlarged cavities or tissue loss) rather than background noise.*

IMAGE PREPROCESSING PIPELINE

The study used a brain MRI dataset containing both baseline and augmented scans distributed across 4 distinct clinical stages:

1. Non-Demented
2. Very Mild Demented
3. Mild Demented
4. Moderate/Severe Demented

Matrix Conversion & Dimensionality Harmonization

Raw image files were converted to numerical NumPy arrays, normalizing pixel intensities and resizing images to $224 \times 224 \times 3$ RGB format to match ResNet-50's pre-trained input specifications.

Data Shuffling & One-Hot Target Encoding

Image arrays were randomly shuffled to prevent the network from learning sequential folder patterns. To avoid imposing an artificial numerical order between stages, clinical labels were mapped into 4-dimensional one-hot vectors, ensuring the network treats each diagnostic phase as a distinct category for multi-class cross-entropy loss calculation.

Test Cohort Balancing & Inverse-Frequency Weighting

Medical imaging datasets typically exhibit severe skew toward specific stages. To handle class imbalance across disease stages, inverse-frequency class weights were incorporated into the loss function during training. This heavily penalized misclassifications on underrepresented stages, preventing the model from biasing predictions toward majority classes.

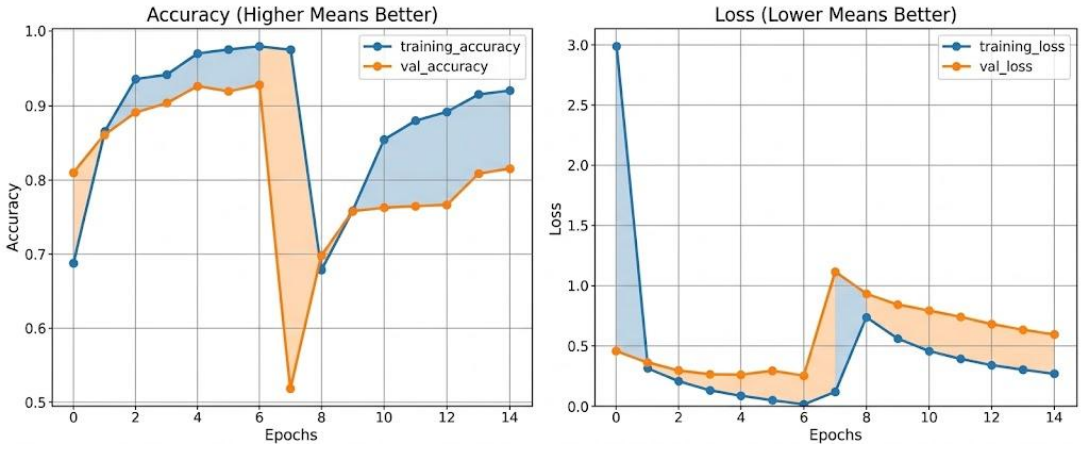
DEEP LEARNING ARCHITECTURE & TRANSFER LEARNING

Transfer Learning Backbone: ResNet-50

Rather than training a randomly initialized network on a limited medical sample, a transfer learning strategy was implemented. After benchmarking multiple backbones (DenseNet, EfficientNet, ResNet-152, InceptionV3), **ResNet-50** was selected for its optimal balance of representational depth and computational efficiency:

- **Residual Skip Connections:** ResNet-50 addresses vanishing gradient degradation across 50 layers via residual identity shortcuts, allowing early edge, contour, and texture filters to propagate directly to deeper semantic layers.
- **Frozen Convolutional Feature Extractor:** The core convolutional layers (pre-trained on ImageNet) were frozen, preserving generic visual filters and reducing optimization overhead.
- **Custom Classification Head:** A Flattening layer was coupled with a Dense fully connected block of 256 neurons, feeding into a 4-unit Dense output layer with Softmax activation to produce calibrated posterior class probabilities.

Hyperparameter Convergence & Early Stopping



Training dynamics were monitored across 15 epochs to establish the exact regularization frontier:

- **Epoch Selection:** As shown in the learning curves, the model converged at **Epoch 5**.
- **Overfitting Boundary:** While training accuracy continued to approach 1.0 at Epoch 6, validation loss spiked sharply and validation accuracy dropped. Epoch 5 was established as the optimal checkpoint, maximizing generalization to unseen patient cohorts while halting over-parameterization.

MODEL EXPLAINABILITY & SALIENCY GRADIENT MAPPING

To inspect the internal decision-making of the network and verify that predictions rely on genuine clinical features, gradient-based saliency maps were computed:

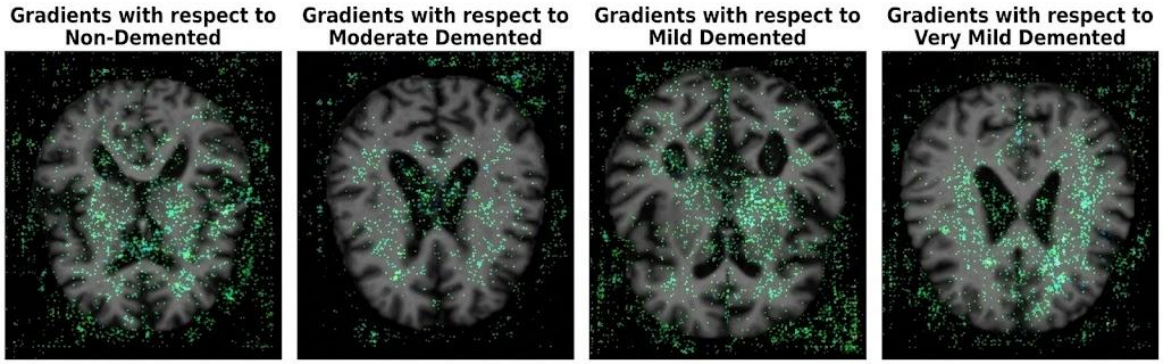
$$\text{Saliency} = \left| \frac{\partial y_c}{\partial X_{i,j}} \right|$$

Where:

- y_c represents the predicted output score (probability) for a given clinical diagnostic class c.
- $X_{i,j}$ is the intensity value of the input MRI pixel at coordinate (i, j).
- $\frac{\partial y_c}{\partial X_{i,j}}$ represents the gradient (partial derivative), which quantifies how sensitive the model's decision is to subtle variations in that specific pixel. A higher gradient indicates an anatomical area with a strong influence on the classification outcome.
- The **absolute value** | … | captures the overall importance magnitude of each pixel, regardless of whether it influences the prediction positively or negatively.

Interpretability:

1. **Brain Boundary Detection:** Across all four stages, the model first maps the outer edges of the skull to gauge overall brain size and scale.
2. **Focus on Central Cavities & Memory Centers:** The strongest heatmaps appear around the fluid-filled cavities in the center of the brain (which expand as brain tissue is lost) and core memory regions, specifically targeting uneven shrinkage between the left and right hemispheres.
3. **Internal Cavities vs. Outer Surface Loss:** While the outer surface of the brain shrinks subtly across the entire head, the model relies far more on the clear enlargement of the central cavities to determine how advanced the disease is.



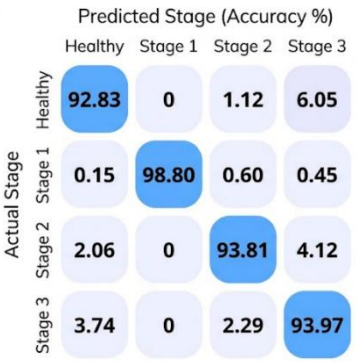
Clinical Takeaway: Saliency mapping confirmed that the network ignores peripheral scan noise and background black space. It first establishes head proportions along the skull contour, then targets the central core of the brain: the ventricles (cavities that expand as surrounding tissue degenerates) and hemispheric asymmetry. The visible enlargement of these deep central cavities provides the model with a clearer diagnostic signal than the subtle, diffuse surface thinning across the outer cortex.

EVALUATION & RESULTS

Multi-Class Staging Performance

Evaluated on an independent test dataset, the model achieved an overall accuracy of **94.33%** across all four clinical categories.

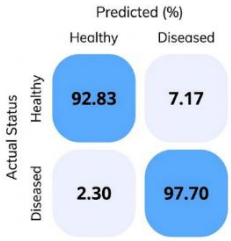
- The model demonstrated high reliability in detecting early-stage neurodegeneration (*Stage 1 / Very Mild*), identifying it correctly in **98.80%** of cases.
- However, diagnostic errors occurred primarily between the two polar extremes rather than adjacent classes:
    - **6.05%** of healthy scans were misclassified as severe dementia (*Stage 3*).
    - **3.74%** of severe dementia scans were misclassified as healthy.
    - This unexpected confusion between opposite ends of the clinical spectrum is directly related to the gradient attention patterns observed in the section above (the model relying heavily on hemispheric asymmetry) and is explored in more detail in the section below.



Binary Pathology Screening (Demented vs. Healthy Cohort)

Considering predictions as a binary classification scheme (healthy vs. diseased) demonstrates the model's suitability as a clinical triage tool:

- **Disease Detection:** 97.70% of all Alzheimer's patients correctly detected.
- **Healthy Identification:** Verified **92.83%** of healthy brain scans.
- **Overall Binary Accuracy:** Achieved **96.16%** diagnostic accuracy in distinguishing pathological from healthy brains.
- **Clinical Error Distribution:** The model produces more false positives (**7.17%**, healthy individuals flagged as diseased) than false negatives (**2.30%**, diseased patients missed). In medical screening, prioritizing a low false negative rate is critical, as it minimizes the risk of leaving an affected patient undiagnosed, while false positives can be re-evaluated during confirmatory follow-up consultations.



THE BIOLOGICAL SYMMETRY PARADOX & CLINICAL WORKFLOW INTEGRATION

The Morphological Paradox

Error analysis revealed that classification confusion did not occur between adjacent stages, but between the two morphological extremes: **healthy brains and advanced dementia brains** (6.05% of healthy brains misclassified as severe; 3.74% of severe brains misclassified as healthy).

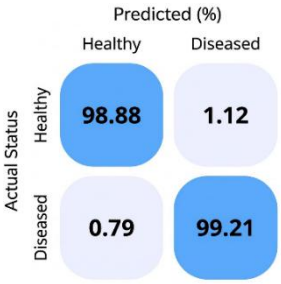
- **The Cause:** Intermediate stages (Very Mild, Mild) exhibit pronounced **hemispheric asymmetry**, as neurodegeneration typically affects one cerebral hemisphere or temporal lobe before the other. The CNN relies heavily on this asymmetry.
- **The Symmetrical Dilemma:** Healthy brains are bilaterally symmetrical. However, in advanced/severe dementia, widespread cortical degeneration and bilateral ventricular dilation restore **bilateral structural symmetry** across the brain. Lacking asymmetry cues, the network must rely on subtle tissue contrast differences, causing occasional confusion.

Multimodal Human-in-the-Loop for Diagnostic Improvement

The ResNet-50 model's primary blind spot was confusing the two structural extremes due to shared bilateral symmetry: **81 healthy scans were labeled as severe**, and **44 severe scans were labeled as healthy** (125 misclassifications total).

In real world clinical practice, differentiating a healthy individual from a patient with severe alzheimer requires only a brief bedside interaction or basic cognitive check. Integrating this basic check to resolve the symmetry issue yields significant performance gains:

- **Percentage of healthy detected patients**
  - **Impact** → from 92.83% to 98.88%
  - **Relevance** → Reclassifying the 81 cases detected wrongly as unhealthy helps reduce false-alarm anxiety for healthy individuals.
- **Percentage of severe detected cases**
  - **Impact** → from 93.97% to 97.71%
  - **Relevance** → Correcting the 44 missed severe cases helps track better advanced patients who need treatment.
- **Percentage of multi-class accuracy**
  - **Impact** → from 94.33% to 97.27%
  - **Relevance** → Correcting those 125 total scans increases overall stage prediction accuracy.
- **Percentage of binary screening accuracy**
  - **Impact** → from 96.16% to 99.11%
  - **Relevance** → Overall pathology detection error drops to under 1%



CONCLUSIONS

This project developed an explainable deep learning computer vision pipeline demonstrating that transfer learning with ResNet-50 can stage Alzheimer's Disease from structural brain MRI with **94.33% multi-class accuracy across all four phases** and **96.16% binary screening accuracy** (detecting **97.70% of all pathological cases**).
By tuning the model to avoid overfitting, balancing class representations with inverse-frequency weighting, and verifying model attention via gradient saliency mapping, the project uncovered the biological symmetry paradox responsible for extreme-class confusion. Resolving this edge-case through a hybrid human-in-the-loop workflow improves the accuracy to **99.11% for disease screening,** minimizing clinical risk of leaving patients undiagnosed, and **97.27% for exact 4-stage classification**. Therefore, this work provides a reliable framework for deploying AI-assisted triage into clinical neurology.

Clinical Diagnostic Metric	ResNet-50	Multimodal check	Impact
Disease detection (binary accuracy)	96.16%	**99.11%**	High precision separation of pathological vs. healthy brains. It's the % of correctness over the entire cohort.
Pathology detection	97.70%	**99.21%**	Correctly identifies 2,888 of 2,911 patients with Alzheimer's. Considers detected pathological over the total pathological.
Healthy identification	92.83%	**98.88%**	Drops healthy false-alarm rate from 7.17% to 1.12%
False negative rate (missed cases)	2.30%	**0.79%**	Minimizes risk of leaving patients undiagnosed
Exact multi-stage accuracy (4 Classes)	94.33%	**97.27%**	Accurately separates progressive stages
Stage 1 (very mild) reliability	98.80%	**98.80%**	Crucial window for early medical intervention
Stage 3 (severe)	93.97%	**97.71%**	Resolves extreme misclassifications into the healthy cohort